

A Frobenius certificate for central-binomial cancellation in Zudilin's symmetric $\zeta(5)$ construction

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Abstract

Brown and Zudilin's totally symmetric cellular approximations to $\zeta(5)$ lead, after eliminating $\zeta(3)$ from Zudilin's two companion hypergeometric forms, to a denominator target $12 d_n^5 P_n \in \mathbb{Z}$ that is equivalent to the central-binomial divisibility $\binom{2n}{n} \mid 6A_n$, where $A_n = 2d_n^5 p_n \in \mathbb{Z}$ is Zudilin's integral numerator. The primes $n < p \leq 2n$ divide $\binom{2n}{n}$ but not d_n , so their cancellation cannot come from any d_n -denominator estimate; it was previously known only experimentally. On this clean interval the target reduces to a single congruence (W): the $\zeta(3)$ -coefficients w_n and $\tilde{w}_n$ of the two forms vanish modulo p . We prove (W) for *all* n by exhibiting an explicit polynomial certificate over $\mathbb{F}_p$. The certificate is $\varphi(k) = (k^p - k)^2$: the Frobenius identity $k^p - k = (k + j)^p - (k + j)$ makes $\varphi \equiv (k + j)^2 \pmod{(k + j)^6}$ at every pole $-j$, while $\deg \varphi = 2p \leq 4n + 2$ keeps φR_n decaying at infinity, so a residue-sum argument on $\mathbb{P}^1$ forces $w_n \equiv 0$; the companion $\tilde{w}_n$ is handled by $\psi(k) = -k(k + n)\varphi(k)$. This closes the entire large-prime (Phase 1) interval unconditionally, correcting Brown–Zudilin's printed inclusion $d_n^5 P_n \in \mathbb{Z}$ (which already fails at $n = 2$) on its critical primes. The remaining primes $p \leq n$ (Phase 2) are discussed but not settled here.

1 Introduction

Let $d_n = \text{lcm}(1, 2, \dots, n)$. Brown and Zudilin [1] study a totally symmetric family of cellular integrals on $\overline{\mathcal{M}}_{0,8}$ producing rational approximations $I'_n = Q_n \zeta(5) - P_n$ to $\zeta(5)$. Their normalized coefficient P_n is Zudilin's numerator p_n divided by a central binomial coefficient,

$$P_n = \frac{(-1)^{n+1} p_n}{\binom{2n}{n}}, \quad (1)$$

where $q_n, p_n, \tilde{p}_n$ are the three solutions of Zudilin's third-order $\zeta(5)$ recurrence [2]. Zudilin's denominator theorem gives $q_n \in \mathbb{Z}$, $2d_n^5 p_n \in \mathbb{Z}$ and $2d_n^3 \tilde{p}_n \in \mathbb{Z}$. Setting

$$A_n := 2 d_n^5 p_n \in \mathbb{Z}, \quad (2)$$

the sharp denominator target for the symmetric family is

$$12 d_n^5 P_n \in \mathbb{Z} \quad (n \geq 1). \quad (\text{T})$$

The multiplier 12 is genuinely necessary: the printed Brown–Zudilin experimental inclusion $d_n^5 P_n \in \mathbb{Z}$ already fails at $n = 2$, where $P_2 = 1190161/384$, $d_2 = 2$, and $d_2^5 P_2 = 1190161/12 \notin \mathbb{Z}$, whereas $12 d_2^5 P_2 = 1190161 \in \mathbb{Z}$. Combining (1)–(2),

$$(\text{T}) \iff \binom{2n}{n} \mid 6A_n \iff \text{ord}_p(A_n) \geq \text{ord}_p\left(\binom{2n}{n}\right) - \text{ord}_p(6) \quad \forall p. \quad (\text{CB})$$

Zudilin's theorem proves only that A_n is an *integer*; the division by $\binom{2n}{n}$ in (1) imposes the additional arithmetic (CB) on the numerator.

The decisive primes are $n < p \leq 2n$. Each such p divides $\binom{2n}{n}$ exactly once, but does *not* divide d_n ; consequently no denominator estimate based on d_n can account for the required cancellation. On this interval (CB) reduces to a single congruence, which we now recall, and then prove.

Setup and the reduction to a congruence

Zudilin's summand is the rational function

$$R_n(k) = (n!)^4 \left(k + \frac{n}{2}\right) \frac{\prod_{j=1}^n (k-j) \prod_{j=1}^n (k+n+j)}{\prod_{j=0}^n (k+j)^6}, \quad (3)$$

whose numerator has degree $2n+1$ and denominator degree $6n+6$, so

$$R_n(k) = O(k^{-4n-5}) \quad (k \rightarrow \infty), \quad \text{i.e.} \quad \text{ord}_\infty R_n = 4n+5. \quad (4)$$

The partial-fraction expansion

$$R_n(k) = \sum_{i=1}^6 \sum_{j=0}^n \frac{a_{i,j}}{(k+j)^i} \quad (5)$$

has coefficients $a_{i,j} \in \mathbb{Q}$, each a $\mathbb{Z}$ -linear combination of $(n!)^4$, $(\frac{n}{2}-j)$ and reciprocals of integers $m-j$ with $|m-j| \leq n$; in particular every $a_{i,j}$ is p -integral for any prime $p > n$. Zudilin's first form and its companion are

$$r_n = \sum_{k \geq 1} R_n(k) = u_n \zeta(5) + w_n \zeta(3) - v_n, \quad \tilde{r}_n = \sum_{k \geq 1} (-k(k+n)R_n(k)) = \tilde{u}_n \zeta(5) + \tilde{w}_n \zeta(3) - \tilde{v}_n,$$

and eliminating $\zeta(3)$ yields $\tilde{w}_n r_n - w_n \tilde{r}_n = q_n \zeta(5) - p_n$ with $p_n = \tilde{w}_n v_n - w_n \tilde{v}_n$. Because $\sum_{k \geq 1} (k+j)^{-i} = \zeta(i) - H_j^{(i)}$ (with $H_j^{(i)} = \sum_{m=1}^j m^{-i}$), the $\zeta(3)$ -coefficients are exactly the pole sums

$$w_n = \sum_{j=0}^n a_{3,j}, \quad \tilde{w}_n = \sum_{j=0}^n \tilde{a}_{3,j}, \quad (6)$$

where $\tilde{a}_{i,j}$ are the partial-fraction coefficients of $-k(k+n)R_n(k)$, given by $\tilde{a}_{i,j} = j(n-j)a_{i,j} + (2j-n)a_{i+1,j} - a_{i+2,j}$ (indices outside $[1,6]$ being zero).

Proposition 1 (Reduction on the clean interval; [3, §1]). *Fix a prime p with $n < p \leq 2n$ and $p \geq 5$. Then $\text{ord}_p \binom{2n}{n} = 1$ and $\text{ord}_p(6) = 0$, and*

$$w_n \equiv 0 \pmod{p} \quad \text{and} \quad \tilde{w}_n \equiv 0 \pmod{p} \quad \implies \quad \text{(CB) holds at } p.$$

Proof. Since $n < p \leq 2n$, exactly one multiple of p lies in $\{n+1, \dots, 2n\}$ and none in $\{1, \dots, n\}$, so $\text{ord}_p \binom{2n}{n} = \text{ord}_p(2n)! - 2\text{ord}_p n! = 1$; and $p \geq 5$ gives $\text{ord}_p 6 = 0$. As p is odd and $p > n$, $\text{ord}_p(2d_n^5) = 0$, whence $\text{ord}_p A_n = \text{ord}_p p_n$. All $a_{i,j}$ and all $H_j^{(i)}$ ($j \leq n < p$) are p -integral, so $v_n = \sum_{i,j} a_{i,j} H_j^{(i)}$ and $\tilde{v}_n$ are p -integral. With $\text{ord}_p w_n, \text{ord}_p \tilde{w}_n \geq 1$ by hypothesis, $\text{ord}_p p_n = \text{ord}_p(\tilde{w}_n v_n - w_n \tilde{v}_n) \geq 1$. $\square$

Thus, on the whole clean interval, (CB) follows from

$$w_n \equiv \tilde{w}_n \equiv 0 \pmod{p} \quad \text{for all primes } n < p \leq 2n, \quad p \geq 5. \quad (\text{W})$$

This note proves (W) for every n .

Theorem 2 (Main result). *For every integer $n \geq 1$ and every prime p with $n < p \leq 2n$ and $p \geq 5$,*

$$w_n \equiv 0 \pmod{p} \quad \text{and} \quad \tilde{w}_n \equiv 0 \pmod{p}.$$

Consequently, by Proposition 1, the central-binomial divisibility (CB) holds at every prime of the clean interval $n < p \leq 2n$, for all n .

The proof is a two-line residue computation once the right polynomial is in hand. Its content is the polynomial itself, and its provenance is the Frobenius endomorphism of $\mathbb{F}_p[k]$. We isolate the mechanism in §2 (a residue extraction valid over any field) and §3 (the certificate), prove Theorem 2 in §4, record the resulting explicit $\mathbb{F}_p$ -certificate and the exact prime window in §5, and state honestly in §6 what remains (the primes $p \leq n$).

2 A residue extraction over an arbitrary field

We work over a field $\mathbb{K}$ and use the residue theorem on $\mathbb{P}^1$: for any $f \in \mathbb{K}(k)$, the sum of the residues of the differential $f dk$ over all points of $\mathbb{P}^1_{\mathbb{K}}$ (the finite poles and the point at infinity) is 0. This holds in every characteristic [4, §II].

Fix distinct points $-0, -1, \dots, -n \in \mathbb{K}$ (automatic when $\text{char } \mathbb{K} > n$) and let

$$G(k) = \sum_{i=1}^6 \sum_{j=0}^n \frac{b_{i,j}}{(k+j)^i} \in \mathbb{K}(k)$$

be any rational function with poles of order ≤ 6 supported on the $-j$.

Lemma 3 (Residue extraction). *Let $\varphi \in \mathbb{K}[k]$ satisfy*

$$\varphi(k) \equiv (k+j)^2 \pmod{(k+j)^6} \quad \text{for every } j = 0, 1, \dots, n. \quad (7)$$

Then for every polynomial $Q \in \mathbb{K}[k]$,

$$\sum_{j=0}^n [QG]_{3,j} = \sum_{j=0}^n \text{Res}_{k=-j}(\varphi(k)Q(k)G(k)),$$

where $[QG]_{3,j}$ denotes the coefficient of $(k+j)^{-3}$ in the partial-fraction expansion of QG . If moreover

$$\text{ord}_{\infty} G - \deg(\varphi Q) \geq 2, \quad \text{equivalently} \quad \deg(\varphi Q) \leq \text{ord}_{\infty} G - 2, \quad (8)$$

then $\varphi QG = O(k^{-2})$ at infinity, its residue there vanishes, and hence

$$\sum_{j=0}^n [QG]_{3,j} = 0.$$

Proof. Q is holomorphic at each $-j$, so the principal part of QG at $-j$ has order ≤ 6 ; write it $\sum_{i=1}^6 [QG]_{i,j}(k+j)^{-i}$. Near $-j$, using (7), the Taylor coefficients of φ are $[(k+j)^m]\varphi = \delta_{m,2}$ for $0 \leq m \leq 5$, so

$$\text{Res}_{k=-j}(\varphi QG) = \sum_{i=1}^6 [QG]_{i,j} [(k+j)^{i-1}]\varphi = \sum_{i=1}^6 [QG]_{i,j} \delta_{i,3} = [QG]_{3,j}.$$

Summing over j gives the first identity. For the second, φQG is a rational function whose only finite poles are the $-j$; the residue theorem gives $\sum_j \text{Res}_{-j}(\varphi QG) + \text{Res}_{\infty}(\varphi QG) = 0$, and (8) makes $\varphi QG = O(k^{-2})$, so $\text{Res}_{\infty}(\varphi QG) = 0$. $\square$

The subtlety that makes the lemma useful is that (7) asks a *single global* polynomial φ to agree with the *local* target $(k+j)^2$ to order 6 at each of $n+1$ points, while keeping $\deg \varphi$ small enough for (8). Over $\mathbb{Q}$ this is impossible for the relevant degree range (the Hermite system is overdetermined). Over $\mathbb{F}_p$ it becomes possible, for exactly the primes we need, by the following observation.

3 The Frobenius certificate

Lemma 4. *Let p be a prime and $a \in \mathbb{F}_p$. In $\mathbb{F}_p[k]$,*

$$k^p - k = (k - a)^p - (k - a).$$

Consequently, for every $a \in \mathbb{F}_p$ and every prime $p \geq 5$, the polynomial $\varphi(k) := (k^p - k)^2$ satisfies

$$\varphi(k) \equiv (k - a)^2 \pmod{(k - a)^6}, \quad \deg \varphi = 2p.$$

Proof. The Frobenius endomorphism of $\mathbb{F}_p[k]$ gives $(k - a)^p = k^p - a^p = k^p - a$, since $a^p = a$ in $\mathbb{F}_p$. Hence $(k - a)^p - (k - a) = k^p - a - k + a = k^p - k$, proving the identity. Writing $u = k - a$, it says $k^p - k = u^p - u$, so

$$\varphi = (u^p - u)^2 = u^2 - 2u^{p+1} + u^{2p}.$$

For $p \geq 5$ we have $p + 1 \geq 6$ and $2p \geq 6$, so $u^{p+1} \equiv u^{2p} \equiv 0 \pmod{u^6}$, leaving $\varphi \equiv u^2 = (k - a)^2 \pmod{(k - a)^6}$. $\square$

Taking $a = -j$ for $j = 0, \dots, n$ shows that a *single* polynomial $\varphi = (k^p - k)^2$ meets the $n + 1$ Hermite conditions (7) simultaneously, with $\deg \varphi = 2p$.

4 Proof of the Main Theorem

Fix $n \geq 1$ and a prime p with $n < p \leq 2n$, $p \geq 5$. All $a_{i,j}$ are p -integral (as $p > n$), so reduce (5) modulo p to obtain $\bar{R}_n(k) = \sum_{i,j} \bar{a}_{i,j} (k + j)^{-i} \in \mathbb{F}_p(k)$; the reductions $\bar{a}_{i,j}$ are the coefficients occurring in (6), and $w_n \equiv \sum_j \bar{a}_{3,j}$, $\tilde{w}_n \equiv \sum_j \bar{a}_{3,j}(\text{companion}) \pmod{p}$. Work over $\mathbb{K} = \mathbb{F}_p$; the points $-0, \dots, -n$ are distinct because $p > n$. Put $\varphi = (k^p - k)^2$, which satisfies (7) by Lemma 4.

The coefficient w_n ($Q = 1$). By Lemma 3 with $G = \bar{R}_n$ and $Q = 1$,

$$w_n \equiv \sum_{j=0}^n \bar{a}_{3,j} = \sum_{j=0}^n \text{Res}_{-j}(\varphi \bar{R}_n) \pmod{p}.$$

Here $\text{ord}_\infty \bar{R}_n = 4n + 5$ and $\deg \varphi = 2p$, so the degree budget (8) reads $\text{ord}_\infty \bar{R}_n - \deg \varphi = (4n + 5) - 2p \geq 2$, i.e. $p \leq 2n + 1$; since $p \leq 2n$ this holds with room to spare and $\varphi \bar{R}_n = O(k^{2p-4n-5}) = O(k^{-5})$. Hence $\text{Res}_\infty(\varphi \bar{R}_n) = 0$ and $w_n \equiv 0 \pmod{p}$.

The coefficient $\tilde{w}_n$ ($Q = -k(k + n)$). The companion form is $-k(k + n)R_n$, so $\tilde{w}_n \equiv \sum_j [QR_n]_{3,j}$ with $Q(k) = -k(k + n)$, $\deg Q = 2$. By Lemma 3,

$$\tilde{w}_n \equiv \sum_{j=0}^n \text{Res}_{-j}(\varphi Q \bar{R}_n) \pmod{p}, \quad \varphi Q \bar{R}_n = O(k^{2p+2-4n-5}).$$

Now $\deg(\varphi Q) = 2p + 2$, so (8) reads $\text{ord}_\infty \bar{R}_n - \deg(\varphi Q) = (4n + 5) - (2p + 2) = 4n + 3 - 2p \geq 2$, i.e. $p \leq 2n$. Under our hypothesis $p \leq 2n$ this holds and $\varphi Q \bar{R}_n = O(k^{-3})$, so again the residue at infinity vanishes and $\tilde{w}_n \equiv 0 \pmod{p}$.

Both congruences hold; with Proposition 1 this proves Theorem 2. $\square$

Remark 5 (The general well-poised weight). The argument used no property of Q beyond its degree. For any $Q \in \mathbb{Z}[k]$ and any prime $p > n$ with $2p \leq 4n + 3 - \deg Q$, the same computation gives

$$\sum_{j=0}^n [QR_n]_{3,j} \equiv 0 \pmod{p}.$$

This is precisely the conjecture (W') of [3, §2] — that the $\zeta(3)$ -coefficient of $\sum_k Q(k)R_n(k)$ vanishes mod p for well-poised weights $Q(-n-k) = Q(k)$ — now proved, and without needing the symmetry of Q : only the degree constraint (equivalently, decay at infinity) is used.

5 The explicit certificate and the prime window

A three-term certificate

The dual of Lemma 3 expresses w_n as an $\mathbb{F}_p$ -linear combination of the exact “decay relations”

$$E_M := \sum_{i=1}^{\min(6,M)} \binom{-i}{M-i} \sum_{j=0}^n j^{M-i} a_{i,j} = 0 \quad (1 \leq M \leq 4n+4),$$

which hold over $\mathbb{Q}$ because they are the vanishing Laurent coefficients $[k^{-M}]R_n = 0$ implied by (4). If $\varphi(k) = \sum_{M \geq 1} c_M k^{M-1}$, then $\sum_M c_M E_M$ is the functional $b \mapsto \sum_j [\varphi\text{-weighted}]$, and the certificate $\varphi = (k^p - k)^2 = k^{2p} - 2k^{p+1} + k^2$ has only three nonzero coefficients:

$$c_3 = 1, \quad c_{p+2} = -2, \quad c_{2p+1} = 1. \quad (9)$$

That is, $w_n \equiv E_3 - 2E_{p+2} + E_{2p+1} \pmod{p}$ as functionals on the partial-fraction data, and each E_M annihilates the true (decaying) vector; the top index $M = 2p+1$ is the minimal truncation that reaches w_n , matching the window bound below. For $\tilde{w}_n$ the certificate is $\psi = -k(k+n)(k^p - k)^2$, of six terms.

Why exactly $n < p \leq 2n$

Two independent constraints pin the interval, and both are visible in the proof.

- *Lower bound $p > n$:* the poles $-0, \dots, -n$ must be distinct in $\mathbb{F}_p$ for the Hermite/residue picture, and $p > n$ makes each $a_{i,j}$ (and each $H_j^{(i)}$) p -integral, so reduction mod p is legitimate.
- *Upper bound $p \leq 2n$ (resp. $2n+1$):* the certificate has degree $2p$, so the decay $\varphi QR_n = O(k^{2p+\deg Q-4n-5})$ requires $2p+\deg Q \leq 4n+3$. For $Q = 1$ this is $p \leq 2n+1$; for $Q = -k(k+n)$ it is $p \leq 2n$. Equivalently, the top functional index is $M = 2p+1$, which must not exceed the available $M \leq 4n+4$: $2p+1 \leq 4n+4 \iff p \leq 2n+1$.

Observation (not needed for the theorem). The upper bound appears sharp for the method: in exact linear-algebra checks for $p > 2n+1$ no polynomial of degree $\leq 4n+3$ meets (7) (the Hermite system is inconsistent mod p), and w_n is there generically a unit. We record this as an observation, verified on small ranges rather than proved; it is not used in Theorem 2, which requires only the *existence* of the certificate on $n < p \leq 2n$, supplied by Lemma 4.

Arithmetic provenance

Lemma 4 is the polynomial shadow of Fermat’s little theorem: the map $k \mapsto k^p - k$ is the additive polynomial vanishing on all of $\mathbb{F}_p$ with derivative -1 , and its being an $\mathbb{F}_p$ -translation invariant of $(k - a)$ is what lets one global object realise $n + 1$ separate local jets. The same mechanism underlies Wolstenholme- and Glaisher-type central-binomial congruences; here it produces the cancellation cleanly and uniformly.

Numerical confirmation

The closed forms $\varphi = (k^p - k)^2$ and $\psi = -k(k + n)(k^p - k)^2$ were checked, via the three-term coefficients (9) and their ψ -analogue, to reproduce the functionals w_n and $\tilde{w}_n$ exactly for all $n \in [2, 60]$ and all primes $n < p \leq 2n$, $p \geq 5$ (420 pairs; zero failures), against the independently computed exact partial-fraction data of [3]. This is a consistency check on the algebra, not the proof: Theorem 2 is unconditional in n .

6 What remains: the primes $p \leq n$

Theorem 2 closes (CB) on the clean interval $n < p \leq 2n$ for all n , hence removes the only primes invisible to d_n -denominator estimates. It does *not* prove the full target (T): the primes $p \leq n$ (Phase 2) remain. There $\text{ord}_p \binom{2n}{n}$ may exceed 1 and interacts with both d_n^5 and the harmonic denominators $H_j^{(i)}$; exact computation through $n = 60$ shows the required inequality (CB) holds but is frequently *tight* (no slack), so a Phase-2 proof must track $\text{ord}_p(p_n)$ against $\text{ord}_p \binom{2n}{n}$ and the harmonic carries simultaneously, rather than deduce it from denominator size. That case is left open. We emphasize the honest scope: what is proved here is the large-prime cancellation (W), unconditionally in n , and with it Phase 1 of (CB); the finite verifications of (T) for small n and the observed sharp 2-adic law are evidence, not theorems.

References

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